DATATALKSCLUB

MLZOOMCAMP 2025

Week 1 notes

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1 Machine Learning (ML)

1.1 What is Machine Learning?

- Is a subfield of Artificial Intelligence, broadly defined as the capability of a machine to imitate intelligent human behaviour.[1]
- Is a type of Artificial Intelligence that allows machines to learn from data without being explicitly programmed. It does this by optimizing model parameters (internal variables) through calculations, such that the model's behaviour reflects the data or experience. The learning algorithm then continously updates the parameter values as learning progresses, enabling the model to learn and make predictions or decisions.[2]

In summary, Machine Learning is a process of extracting patterns from data, you have:

- features: information or attributes of an object.
- target: property or attribute to find out for unseen objects.

New feature values are presented to the model and it makes predictions from the learned patterns.[3]

2 Supervised Machine Learning (SML)

2.1 What is Supervised Machine Learning?

In SML there are always labels associated with features, the model is trained and then it can make predictions on new features. The model is taught by certain features and targets.[4]

- Feature Matrix (X): Made of observations and features, rows and columns respectively.
- Target Variable (y): A vector with the target information you want to predict. For each row of X there is a value in y.

2.2 What about the model?

The model can be presented as a function 'g' that takes 'X' as a parameter and makes predictions as close as possible to the targets in 'y', where:

- g: Is the model.
- X: The feature matrix.
- y: the target values.

Obtaining g is called training.

$$\text{g(X)} \sim y \text{ (1)}$$

Figure 1: *Training a model, example equation.*

2.3 Supervised Machine Learning problems

- **Regression**: Output is a number.
- **Classification**: Output is a category.
  - **Binary**: Two categories.
  - **Multiclass**: More than two categories.
- **Ranking**: Output are top scores associated with corresponding items.

SML teaches a model using different examples to come up with a function using a feature matrix and make predictions as close as possible to the y targets.

3 Cross-Industry Standard Process for Data Mining (CRISP-DM)

Is an open standard process model that describes common approaches used by data mining experts. It is the most widely-used analytics model.[5]

3.1 CRISP-DM Process

- Business understanding:
  - Do you need ML for the project?
  - Project goal has to be measurable.
- Data understanding:
  - Are your data sources reliable?
  - Do you need more data?
  - What data is suitable to measure your goal?
- Data preparation:
  - Clean: Fix datatypes, handle missing, duplicate and outlier values, those are few steps from a wide range of options.
  - Format: Ensure data consistency and validate accuracy.
  - Organize: Sort data, reset index, structure your data in a tabular format.
- Modeling:
  - Train different models, compare your results, decide if it is required to add new features or fix data issues.
- Evaluation:
  - Measure the performance of selected model, Does it accomplish your goal?
- Deployment:
  - Make it available to all users, evaluate your deployment consider metrics to measure model performance and check if data can be improved.

3.2 Maintainability

As in other project it is important to consider how maintainable the project is.

3.2.1 Iteration

ML projects require iterations. Start simple, learn from feedback (evaluation-deployment), improve (iterate) the improvement may have different steps depending on the methodology but they all have similarities and a common objective that is selecting the best performing model. Adopting best practices, a methodology or a systematic process to build a ML project can help to make the whole process more comprehensive and reliable as described by CRISP-DM.[5]

4 Choosing and training a model

Models are fitted with training data used to predict the y values of the validation feature matrix. The validation dataset is not used in training. Then the predicted y values are compared with the actual y values.
The context or nature of your goal yields the model selection, more than one model can help accomplish your goal but only one suits what you want to accomplish.

5 Multiple Comparisons problem (MCP)

By chance any model can be 'lucky' and obtain close predictions because all models are probabilistic. To obtain the best model use a training and validation datasets then use a test dataset to verify if proposed model is suitable to accomplish your goal.
Once the best model is selected the validation and training datasets can be combined to form a single dataset before testing using the test set.[6]

6 Model Selection

The most common approach involves:

- Goal: Analyzing the goal that you want to accomplish.
- Problem type: Such as classification, regression, clustering, etc.
- Data: Quality, feature types, distribution.
- Simple baseline: Start with simple models to establish a performance baseline.
- Try multiple models: Tesiting a range of candidate models including tuning of the models.

The selected model comes after training different models and comparing them, do notice that models can have assumptions or not. In order to accomplish your goal it is important that you are able to identify data-specific factors and discard models that simply are not a good fit for your data or use appropriate techniques to increase performance when possible.

7 Setting up an environment

You have different options to choose from to set up an environment, examples:

- **Python Virtual Environments**
- **Jupyter Notebooks**
- **Colab**

I used Python virtual environments as you have control over the environment and is 'isolated' from the rest of the system.

7.1 Python Virtual Environment

To set a python virtual environment in Pop OS! I did:

- **1**: Install python3.
  - *sudo apt update*
  - *sudo apt install python3*
- **2**: Verify installation, open Terminal and do:
  - *python3 --version*
  - You may see something similar to *Python 3.10.12*
- **3**: Create virtual environment:
  - Go to desired folder location, open Terminal and do:
    - *mkdir environment_container_name*
  - Go inside of previously created folder:
    - *cd environment_container_name*
  - Now create a virtual environment:
  -
    - *python3 -m environment_name*
- **4**: Activate virtual environment:
  - Go inside of previously created virtual environment:
    - *cd environment_name*
  - Activate created virtual environment, do:
    - *source bin/activate*
  - Notice how terminal changes to something like:
    - *(environment_name) user@somewhere:/some/path$*

- **5**: Upgrade pip:
  - To upgrade pip, do:
    - *pip install --upgrade pip*
  - Reading stdout in Terminal you can see what version was installed.
- **6**: Install additional packages, examples:
  - *pip install pandas*
  - *pip install matplotlib*
  - *pip install seaborn*
  - You only have to search for the specific package name you want to install.
  - **Tip**: Once in a while do *pip cache purge* so that you are able to install updated versions of your installed packages and not cached ones (old). Remember how pip was updated, it works in the same way for other packages.
- **7**: Update installed packages, examples:
  - *pip install --upgrade package_name*
    - *pip install --upgrade pandas*
    - *pip install --upgrade matplotlib*
    - *pip install --upgrade seaborn*
- **8**: Open the environment in a code editor such as VSCode. If you want to use notebooks, you must activate the environment first and launch VSCode from your created environment location i.e. environment_container_name/environment_name/ and choose/install a python kernel to be able to execute python code in notebook code cells. Make sure that you have all your packages installed to prevent importing errors.

8 Numpy

I am checking this handbook [7] it covers numpy, pandas, matplotlib and some ML.

9 Linear Algebra Refresher

I am checking:

- Linear Algebra playlist [8]
- Linear Algebra playlist [9]
- Linear Algebra Notebook [10]

I have been going through some videos and the notebook, this takes time, I find the notebook amazing, wish I had the chance to learn this way when I was a young student.

References

[1] Sara Brown. ”Machine learning, explained” [Online], at: What is machine learning? [Consulted on September 2025].

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[3] DataTalksClub (github). ”Introduction to Machine Learning” [Online], at: Notes [Consulted on September 2025].

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[5] DataTalksClub (github). ”CRISP-DM” [Online], at: Notes [Consulted on September 2025].

[6] DataTalksClub (github). ”Model Selection” [Online], at: Notes [Consulted on September 2025].

[7] jakevdp (github). ”PythonDataScienceHandbook” [Online], at: PythonDataScienceHandbook [Consulted on September 2025].

[8] Dr Linda Nerd’s Lesson (Youtube Channel). ”Linear Algebra” [Online], at: LinearAlgebra1 [Consulted on September 2025].

[9] Adrian Banner OCS (Youtube Channel). ”Linear Algebra” [Online], at: LinearAlgebra2 [Consulted on September 2025].

[10] weijie-chen (github). ”Linear Algebra” [Online], at: LinearAlgebra3 [Consulted on September 2025].